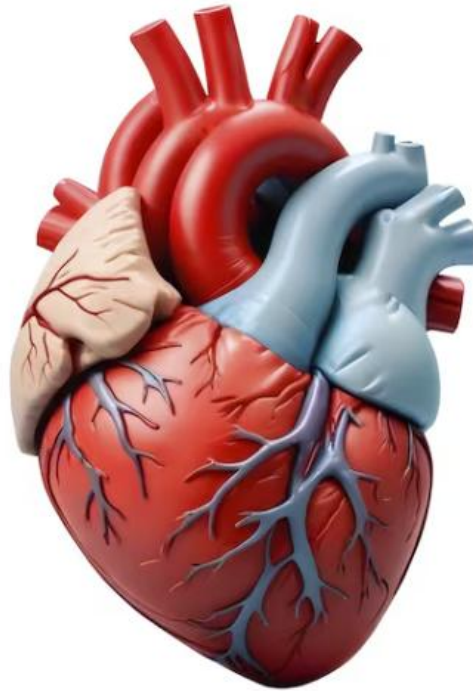


# ANALYZING REPORT

Predicting Cardiovascular Disease



❑ Name: Huzaifa Baig

❑ Dataset : Predicting Cardiovascular Disease

❑ Assigned by : Sir Yasir Nawaz

# INTRODUCTION

- This heart disease dataset is common features which makes it one of the heart disease dataset available so far for research purposes.
- This dataset consists of 1000 subjects with 14 features.
- This dataset will be useful for building an early-stage heart disease detection as well as to generate predictive machine learning models.

# Cardiovascular Disease Dataset Description

| S.No | Attribute                         | Assigned Code     | Unit                                                                                                                                                                                                                           | Type of the Data |
|------|-----------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1    | Patient Identification Number     | patientid         | Number                                                                                                                                                                                                                         | Numeric          |
| 2    | Age                               | age               | In Years                                                                                                                                                                                                                       | Numeric          |
| 3    | Gender                            | gender            | 1,0(0= female, 1 = male)                                                                                                                                                                                                       | Binary           |
| 4    | Chest pain type                   | chestpain         | 0,1,2,3<br>(Value 0: typical angina<br>Value 1: atypical angina<br>Value 2: non-anginal pain<br>Value 3: asymptomatic)                                                                                                         | Nominal          |
| 5    | Resting blood pressure            | restingBP         | 94-200 (in mm HG)                                                                                                                                                                                                              | Numeric          |
| 6    | Serum cholesterol                 | serumcholesterol  | 126-564 ( in mg/dl)                                                                                                                                                                                                            | Numeric          |
| 7    | Fasting blood sugar               | fastingbloodsugar | 0,1 > 120 mg/dl<br>(0 = false, 1 = true)                                                                                                                                                                                       | Binary           |
| 8    | Resting electrocardiogram results | restingelectro    | 0,1,2<br>(Value 0: normal,<br>Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV), Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria) | Nominal          |

# Cardiovascular Disease Dataset Description

|    |                                       |                  |                                                                 |         |
|----|---------------------------------------|------------------|-----------------------------------------------------------------|---------|
| 9  | Maximum heart rate achieved           | maxheartrate     | 71-202                                                          | Numeric |
| 10 | Exercise induced angina               | exerciseangia    | 0,1 (0 = no, 1 = yes)                                           | Binary  |
| 11 | Oldpeak =ST                           | oldpeak          | 0-6.2                                                           | Numeric |
| 12 | Slope of the peak exercise ST segment | slope            | 1,2,3<br>(1-upsloping, 2-flat, 3-downsloping)                   | Nominal |
| 13 | Number of major vessels               | noofmajorvessels | 0,1,2,3                                                         | Numeric |
| 14 | Classification                        | target           | 0,1 (0= Absence of Heart Disease, 1= Presence of Heart Disease) | Binary  |

# Cardiovascular Disease Actual Dataset

| age | gender | chestpain | restingBP | serumcholesterol | fastingbloodsugar | restingelectro | maxheartrate | exerciseangia | oldpeak | slope | noofmajorvessels | target |
|-----|--------|-----------|-----------|------------------|-------------------|----------------|--------------|---------------|---------|-------|------------------|--------|
| 53  | 1      | 2         | 171       | 0                | 0                 | 1              | 147          | 0             | 5.3     | 3     | 3                | 1      |
| 40  | 1      | 0         | 94        | 229              | 0                 | 1              | 115          | 0             | 3.7     | 1     | 1                | 0      |
| 49  | 1      | 2         | 133       | 142              | 0                 | 0              | 202          | 1             | 5.0     | 1     | 0                | 0      |
| 43  | 1      | 0         | 138       | 295              | 1                 | 1              | 153          | 0             | 3.2     | 2     | 2                | 1      |
| 31  | 1      | 1         | 199       | 0                | 0                 | 2              | 136          | 0             | 5.3     | 3     | 2                | 1      |

# Exploratory Data Analysis (EDA)

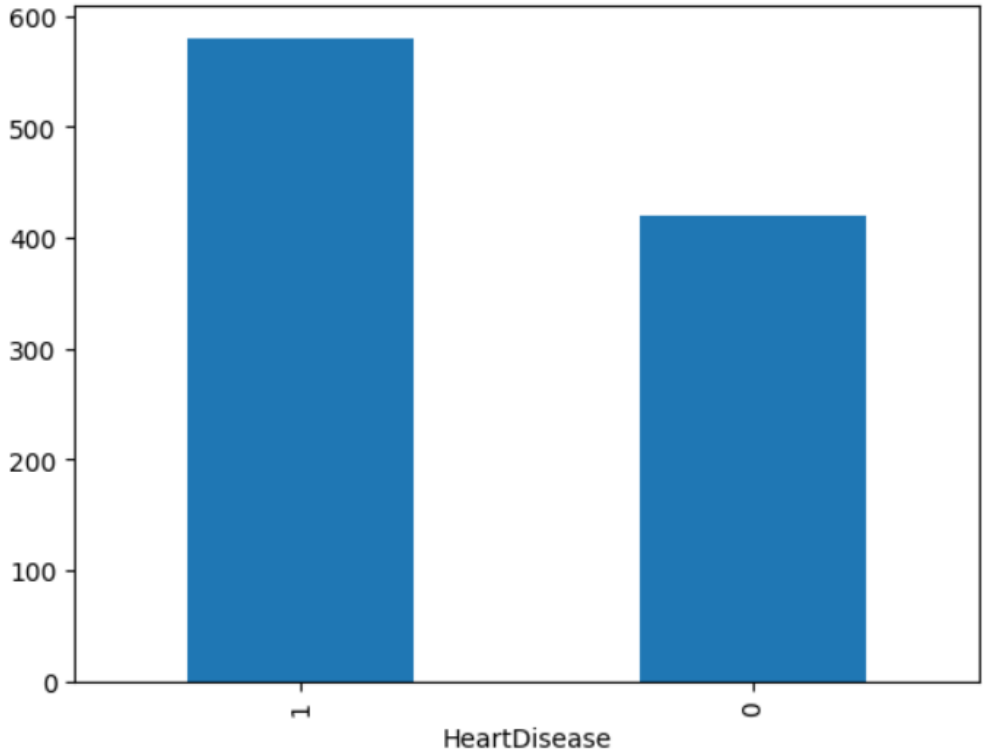
Exploratory Data Analysis (EDA) is an analytical approach to understand a dataset's main characteristics by summarizing and visualizing it, primarily using statistical graphics and data visualization methods to uncover patterns, anomalies, and relationships.

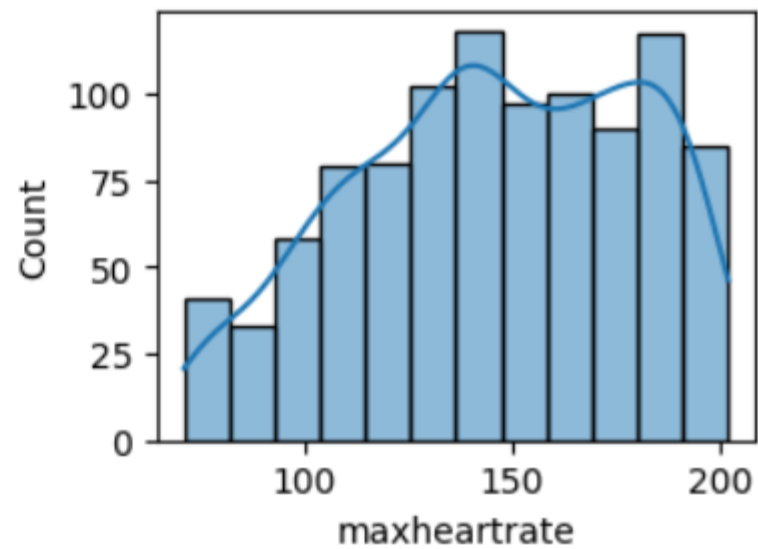
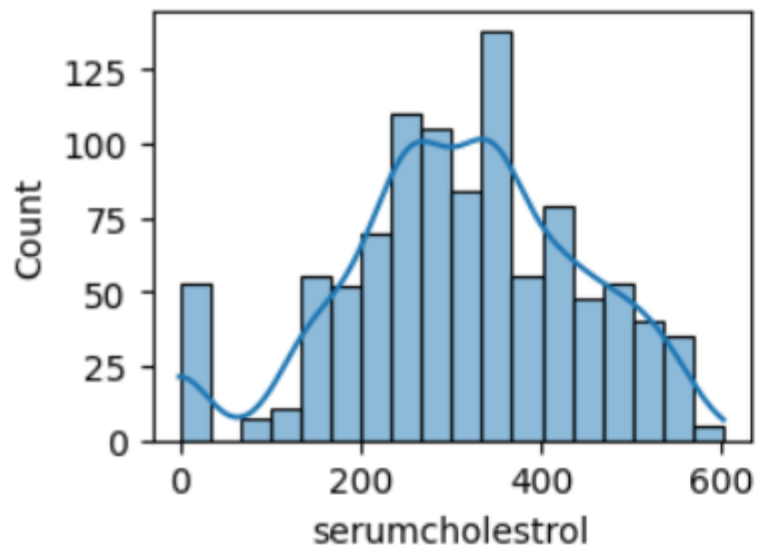
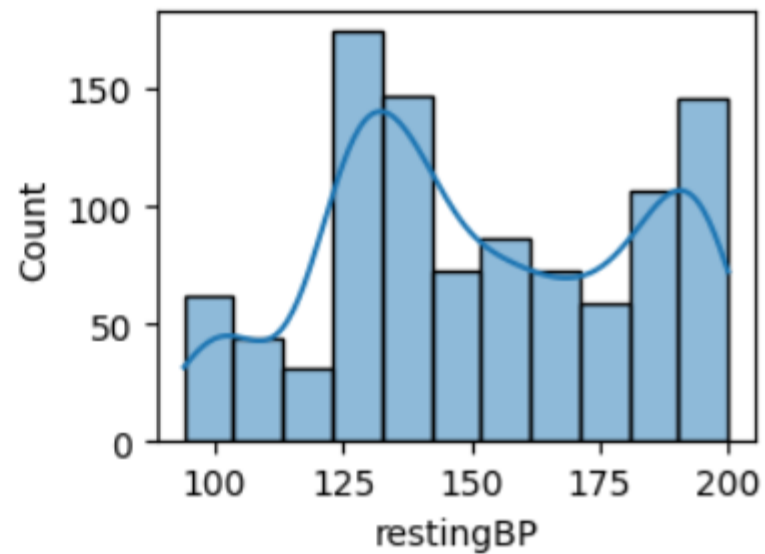
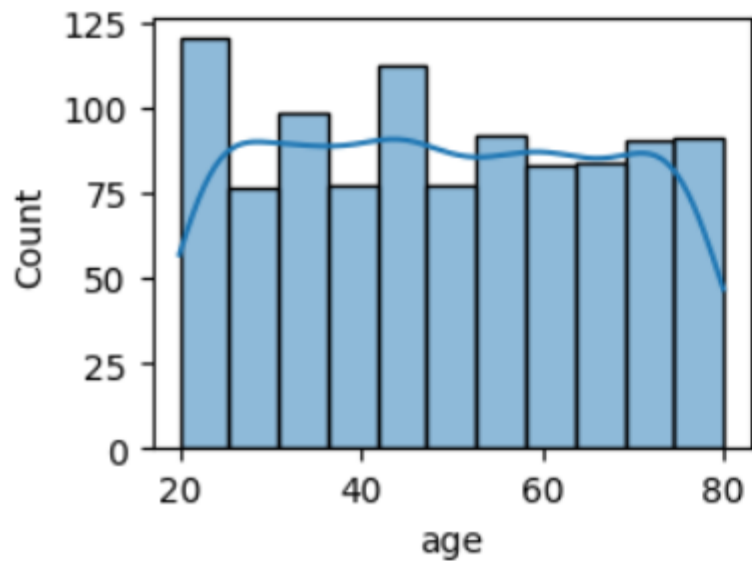
| #  | <i>Columns</i>    | <i>Non-Null</i> | <i>Count</i> | <i>Dtype</i> |
|----|-------------------|-----------------|--------------|--------------|
| 1  | patientid         | 1000            | non-null     | int64        |
| 2  | age               | 1000            | non-null     | int64        |
| 3  | gender            | 1000            | non-null     | int64        |
| 4  | chestpain         | 1000            | non-null     | int64        |
| 5  | restingBP         | 1000            | non-null     | int64        |
| 6  | serumcholesterol  | 1000            | non-null     | int64        |
| 7  | fastingbloodsugar | 1000            | non-null     | int64        |
| 8  | restingrelectro   | 1000            | non-null     | int64        |
| 9  | maxheartrate      | 1000            | non-null     | int64        |
| 10 | exerciseangia     | 1000            | non-null     | int64        |
| 11 | oldpeak           | 1000            | non-null     | float64      |
| 12 | slope             | 1000            | non-null     | int64        |
| 13 | noofmajorvessels  | 1000            | non-null     | int64        |
| 14 | target            | 1000            | non-null     | int64        |

dtypes: float64(1), int64(13)

|                   |                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Columns</b>    | 'patientid', 'age', 'gender', 'chestpain', 'restingBP', 'serumcholesterol', 'fastingbloodsugar', 'restingelectro', 'maxheartrate', 'exerciseangia', 'oldpeak', 'slope', 'noofmajorvessels', 'target' |
| <b>Shape</b>      | (1000, 14)                                                                                                                                                                                           |
| <b>Duplicated</b> | 0                                                                                                                                                                                                    |
| <b>Null</b>       | 0                                                                                                                                                                                                    |

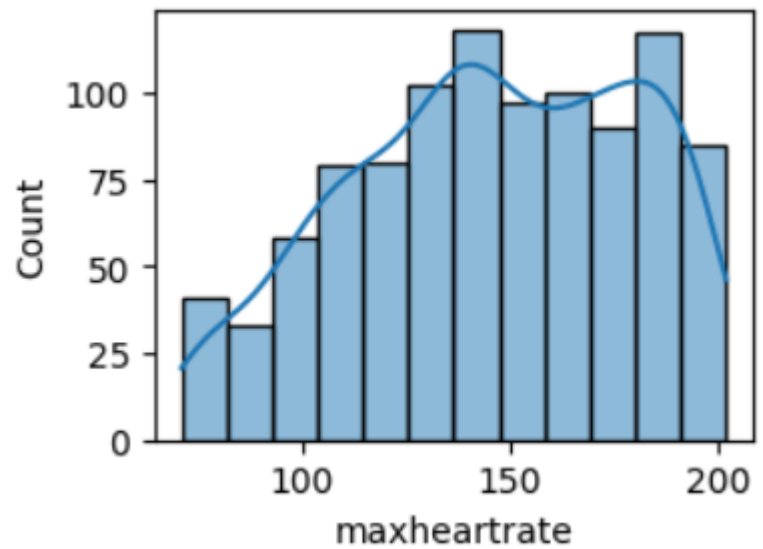
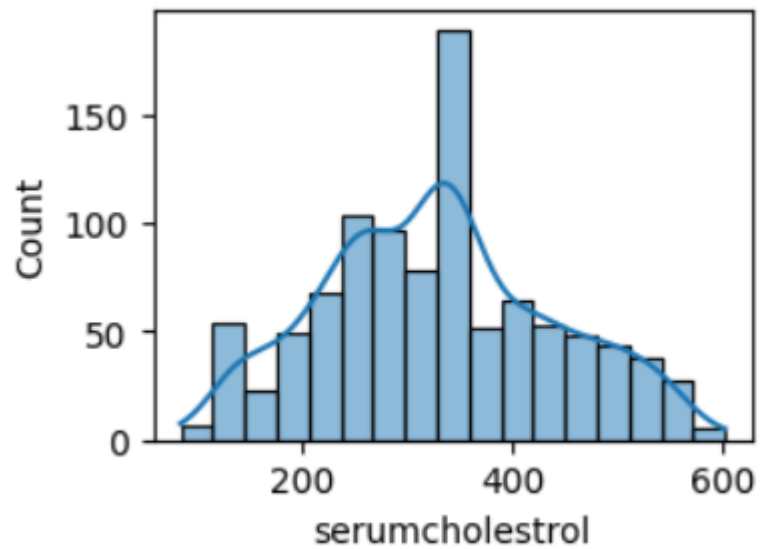
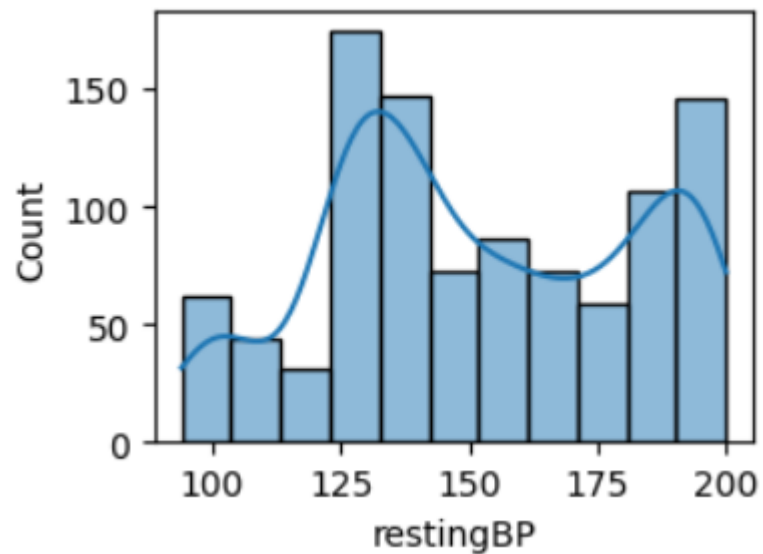
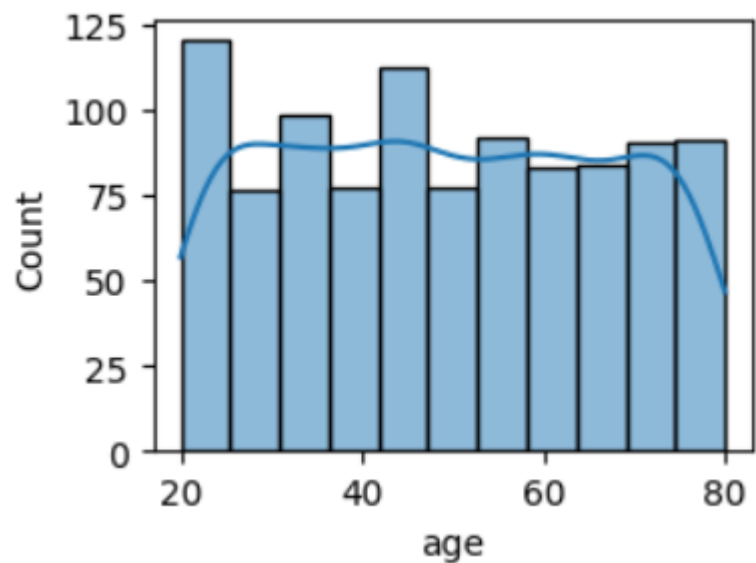
|       | age        | gender      | chestpain   | restingBP   | serumcholesterol | fastingbloodsugar | restingrelectro | maxheartrate | exerciseangia | oldpeak     | slope       |
|-------|------------|-------------|-------------|-------------|------------------|-------------------|-----------------|--------------|---------------|-------------|-------------|
| count | 1000.00000 | 1000.000000 | 1000.000000 | 1000.000000 | 1000.000000      | 1000.000000       | 1000.000000     | 1000.000000  | 1000.000000   | 1000.000000 | 1000.000000 |
| mean  | 49.24200   | 0.765000    | 0.980000    | 151.747000  | 328.877640       | 0.296000          | 0.748000        | 145.477000   | 0.498000      | 2.707700    | 1.878400    |
| std   | 17.86473   | 0.424211    | 0.953157    | 29.965228   | 110.032936       | 0.456719          | 0.770123        | 34.190268    | 0.500246      | 1.720753    | 0.697346    |
| min   | 20.00000   | 0.000000    | 0.000000    | 94.000000   | 85.000000        | 0.000000          | 0.000000        | 71.000000    | 0.000000      | 0.000000    | 1.000000    |
| 25%   | 34.00000   | 1.000000    | 0.000000    | 129.000000  | 249.000000       | 0.000000          | 0.000000        | 119.750000   | 0.000000      | 1.300000    | 1.000000    |
| 50%   | 49.00000   | 1.000000    | 1.000000    | 147.000000  | 328.880000       | 0.000000          | 1.000000        | 146.000000   | 0.000000      | 2.400000    | 2.000000    |
| 75%   | 64.25000   | 1.000000    | 2.000000    | 181.000000  | 404.250000       | 1.000000          | 1.000000        | 175.000000   | 1.000000      | 4.100000    | 2.000000    |
| max   | 80.00000   | 1.000000    | 3.000000    | 200.000000  | 602.000000       | 1.000000          | 2.000000        | 202.000000   | 1.000000      | 6.200000    | 3.000000    |

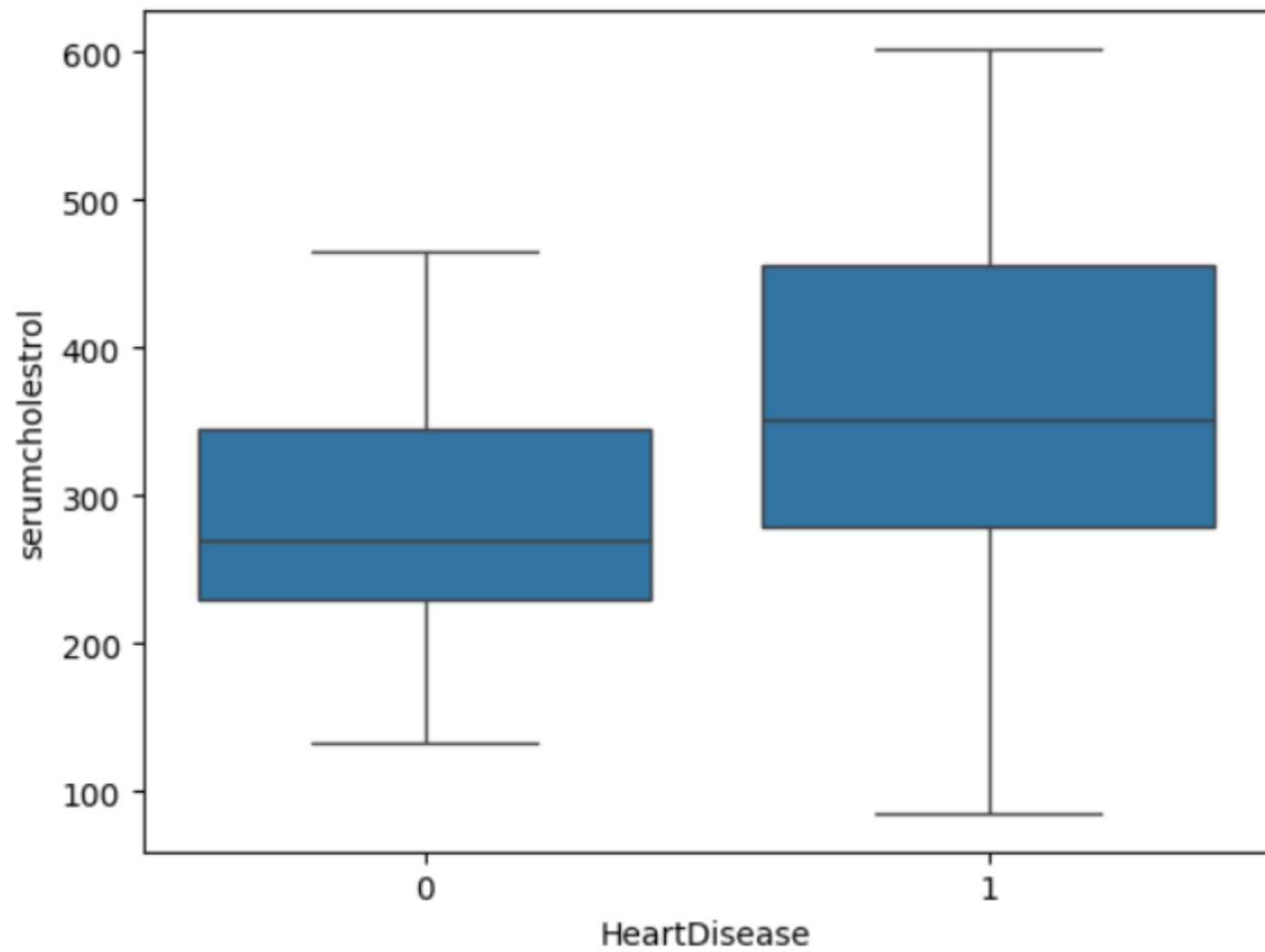




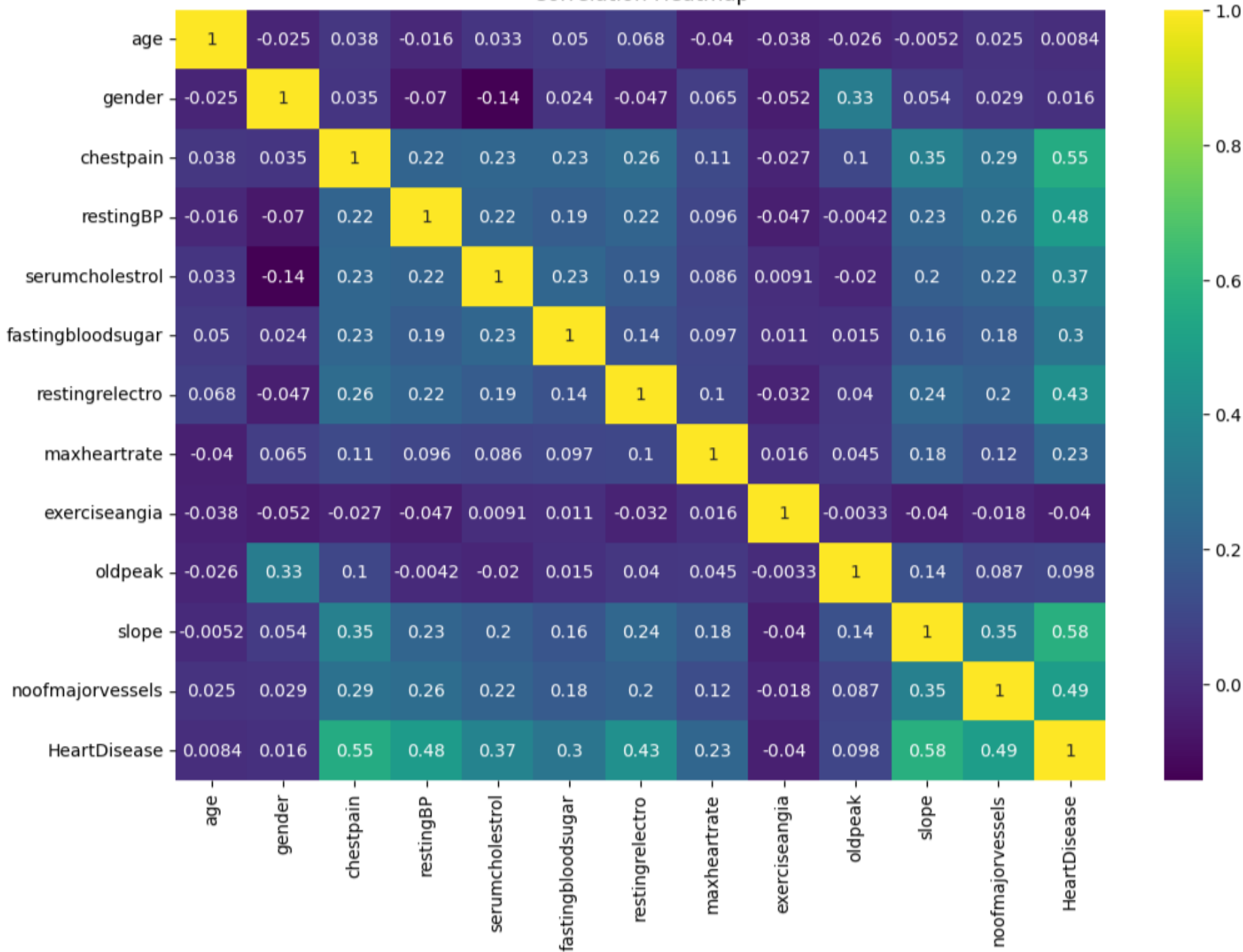
# Data cleaning

Data cleaning, or data cleansing, in machine learning is the process of identifying and rectifying errors, inconsistencies, and inaccuracies within a dataset to improve its quality, reliability, and suitability for training machine learning models. This crucial preprocessing step ensures that the data is accurate, consistent, and free of issues that could negatively impact model performance and lead to unreliable predictions.





Correlation Heatmap



# Apply Machine Learning Algorithm

## Logistic Regression

- Training accuracy: 92.62
- Testing accuracy: 92.50
- Classification Report.

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.93      | 0.89   | 0.91     | 84      |
| 1            | 0.92      | 0.95   | 0.94     | 116     |
| accuracy     |           |        | 0.93     | 200     |
| macro avg    | 0.93      | 0.92   | 0.92     | 200     |
| weighted avg | 0.93      | 0.93   | 0.92     | 200     |

# **Application Interface**

# Heart Stroke Prediction

Provide the following details to check your heart stroke risk:

Age



Gender

M

Chest Pain Type

TA

Resting Blood Pressure (mm Hg)

120 - +

Cholesterol (mg/dL)

200 - +

Fasting Blood Sugar > 120 mg/dL

0

Resting ECG

Normal

Max Heart Rate



Exercise-Induced Angina

0

Oldpeak (ST Depression)



ST Slope

Upsloping

Number of Major Vessels

0

Predict

**THANK YOU**